

A Worked Solution to a Question I Asked on Math StackExchange 5 Years Ago

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1 Introduction

In October of 2021 I asked a [question](#) on [Math.StackExchange](#). I wanted to know if there was a solution to a tricky integral I'd stumbled into.

$$\int \sqrt{17 + 8 \cos(3t) + 9 \cos^2(3t)} dt \tag{1}$$

Within a week or so, a user by the name of [Parcly Taxel](#) had posted a solution. This was wonderful, but I wanted to understand the method of solution myself, not merely use the answer. This proved much more difficult than I had anticipated, and while I was able to follow the first few steps of the solution I was quickly lost. I found myself confused by the application of Möbius Transforms and Elliptic Integrals. Lost in a sea of algebra, I put the problem aside and spent some time improving my Complex Analysis and investigating some strange special functions which my undergraduate electrical engineering education had not introduced me to. Time pressures of work and other hobby projects meant that this goal soon fell by the wayside.

However I've recently found time to pick this project back up! This document contains a fully worked solution and a large amount of context for the work that Parcly Taxel posted. While I wrote it to confirm and solidify my own understanding, I hope it will be of use, or of interest to someone on the internet. The document is written in a conversational style and should be understandable to someone who has completed the usual calculus undergrad sequence along with at least a general introduction to complex numbers.

I've erred on the side of over detailing the mathematics. Having had to parse plenty of difficult to read math texts I'd prefer to say too much rather than too little. I hope that the steps in the math are small enough to easily follow. As we all know, a lot of what passes for mathematical writing is actually just organized notes. Well informed readers can simply skip the parts they know. I should also mention, the topics covered here are not pedagogically efficient. Their selection is based on what I knew and did not know at the time I was working this problem. As is the case with all math, the interested reader will get much more out of this text if they follow along with paper and pencil. As the great Donald Knuth says "Mathematics is a participatory sport; you don't learn by watching it being done, you learn by doing it."

A bibliography of traditional sources is included at the end, however, I should note that I made good use of Google's AI, Gemini 3.0, while putting this work together. Either to help explain context, or double check work I wasn't confident about, or point the way forward if I was lost. This document is my own work, I wrote it myself, but I used AI to help produce it. I am confident in its accuracy, but I am not a professional mathematician, if you find a mistake in the document please let me know and I'll make an

honest effort to correct it. Finally, I've included plenty of hyperlinks where I felt it appropriate, but with the current state of the internet, all I can guarantee is that those links work as of the date I posted this document, April 19, 2026. With those caveats and the stage set, we move forward!

2 A Brick Wall

The integral in (1) shows up when you try and find the arclength of a [trefoil knot](#) in $\mathbb{R}^3$. A knot is a simple closed curve that is embedded in euclidean space, $\mathbb{R}^3$. Here's what a trefoil knot can look like in $\mathbb{R}^3$.



The [arclength](#) of a trefoil knot is the distance the knot traces out. Let's recall the general formula for an arclength, L , in $\mathbb{R}^3$ for euclidean coordinates.

$$L = \int \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt \quad (2)$$

Here, the various $f'(t)$ functions are the derivatives of the parametric function for that coordinate. A parametric equation in $\mathbb{R}^3$ will have 3 equations for the standard x, y and z euclidean coordinates. We'll use one of the standard parametrizations for the trefoil knot. This set of equations trace out the knot as we increment t . Here are the equations that make it up.

$$x = \sin(t) + 2\sin(2t) \quad (3)$$

$$y = \cos(t) - 2\cos(2t) \quad (4)$$

$$z = \sin(3t) \quad (5)$$

Taking the derivative of each of these equations, and putting them in to our arclength equation (2) gives us the integral (1) that started off this whole thing. I'm not going to run through the calculus, as the derivatives of all of these equations are well known and the algebra to simplify things is not complicated. There's also a worked solution [here](#).

You quickly run into a problem when you try and solve the integral though. The standard integral tables you're introduced to in university are of no help and trying some of the tricks you're taught to reformulate things opens no avenues. There is one move that you can do, and that's a change of variable. Let $\phi = 3t$, then $d\phi = 3 dt$ and $\frac{d\phi}{3} = dt$. The substitution is pretty simple and gives us the following.

$$\frac{1}{3} \int \sqrt{17 + 8\cos(\phi) + 9\cos^2(\phi)} d\phi \quad (6)$$

The astute reader will note that there's no real change in the integral. We can go up a level of abstraction and attempt substituting a function rather than a variable. This time we'll let $u = \cos(\phi)$, then $du = -\sin(\phi) d\phi$ and $d\phi = \frac{du}{-\sin(\phi)}$. We can pull the negative out front since it's a constant and if we recall the trigonometric identity $1 = \cos^2(\phi) + \sin^2(\phi)$ we should see that $\sin(\phi) = \sqrt{1 - \cos^2(\phi)}$. We can now complete our u -substitution and obtain the following.

$$-\frac{1}{3} \int \frac{\sqrt{17+8u+9u^2}}{\sqrt{1-u^2}} du \quad (7)$$

Now we have something that at least looks a little different from our starting point. However we're still stuck, our undergraduate integration techniques are of no use here.

3 Elliptic Functions and Elliptic Integrals

The integral I was trying to solve is an *elliptic integral*. An elliptic integral has the form $\int R(u, \sqrt{p(u)}) du$. Here R is a rational function and $p(u)$ is a polynomial of third or fourth degree. With a simple trick shown below, we can see how our integral meets that definition!

$$-\frac{1}{3} \int \frac{\sqrt{17+8u+9u^2}}{\sqrt{1-u^2}} du = -\frac{1}{3} \int \frac{17+8u+9u^2}{\sqrt{1-u^2}\sqrt{17+8u+9u^2}} du \quad (8)$$

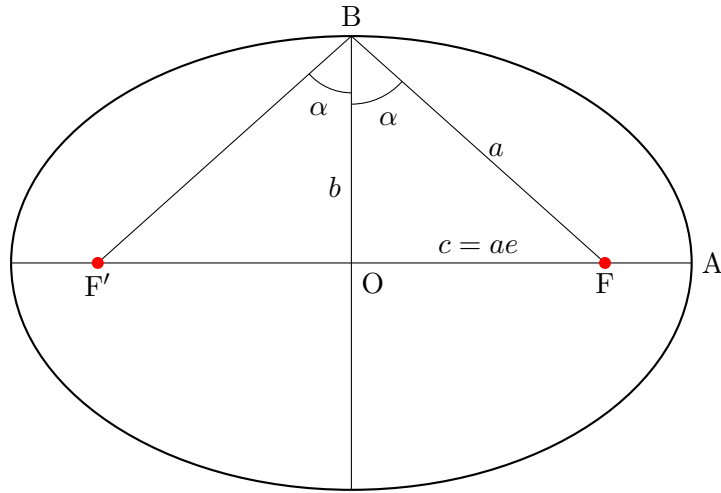
All that's happened is we've multiplied our fraction by $\frac{\sqrt{17+8u+9u^2}}{\sqrt{17+8u+9u^2}}$. Then that gives a rational function, $R(u, \sqrt{p(u)})$, where $p(u) = (1-u^2)(17+8u+9u^2)$. This brings us to the first topic I had no idea about when I asked my initial Math.StackExchange question, special functions! Broadly, a [special function](#) is a *named* function.

“Nommer c'est posséder”

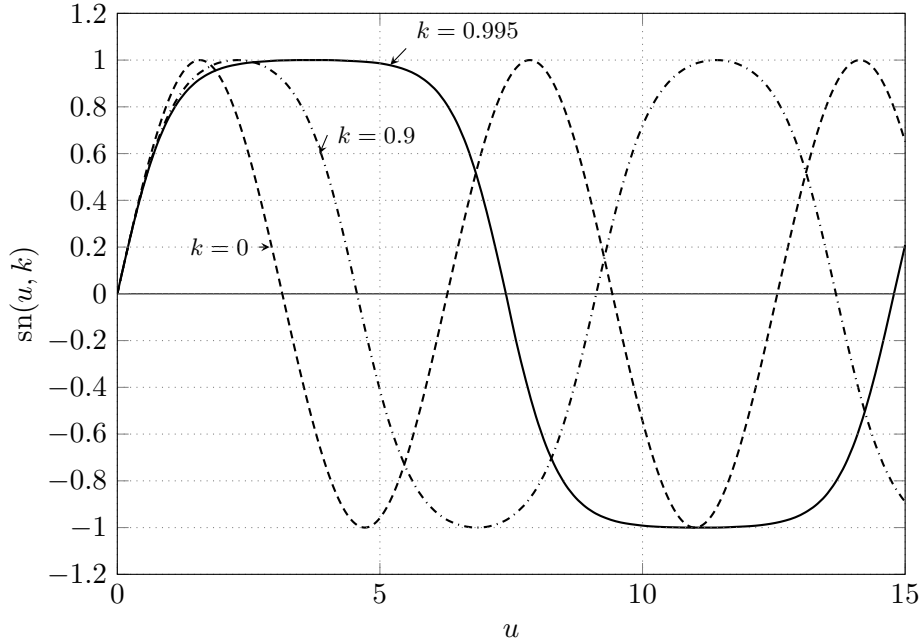
— French Proverb

These functions, like the Bessel or Gamma function were discovered as solutions to specific, (often physics related) problems but were found to have more general uses. One class of special functions are known as elliptic functions. Elliptic functions can be thought of as *generalisations* of trigonometric functions. Let's meet two of them, $\text{sn}(u, k)$ and $\text{cn}(u, k)$, the *elliptic sine* and *elliptic cosine* respectively. We should immediately see that unlike $\sin(\theta)$ or $\cos(\theta)$, they take two inputs, the argument, u , and the modulus, k . While trigonometric functions have a deep relationship with the unit circle, elliptic functions are related to the ellipse (the clue was in the name!). k , the modulus, controls the *eccentricity* of the ellipse. In formal terms, we say

$k = \sin(\alpha)$. The diagram below should make the relationship clear. F and F' are the *focal points* of the ellipse, while α is the *angular eccentricity*. As the focal points move apart, the angular eccentricity of the ellipse increases as does the modulus. Finally, for completeness, c is the *linear eccentricity* of the ellipse, but is not used in the equations above.



The plot below shows the $\text{sn}(u, k)$ function for various values of k . The reader should note that when $k = 0$, it is *identical* to the $\sin(u)$ function. To be precise $\text{sn}(u, 0) = \sin(u)$, where u is a standard input in radians. The generalisation of elliptic functions from trigonometric functions should be readily apparent. As k decreases, our ellipse approaches a circle and so our elliptical functions begin to behave like trigonometric ones. Our argument u operates just like an argument in the classical trigonometric functions, controlling how far around our ellipse we rotate.



John Stillwell has an excellent discussion of elliptic integrals and functions in his book, *Mathematics and Its History*. Right now, for us, the key fact is as follows, any elliptic integral can be written as a linear combination of elementary functions and three special functions, F , E and Π . They are known as the first (F), second (E) and third (Π) kinds of elliptic integrals. There are a number of different formulations of these integrals and functions, the classical form was given by [Legendre](#) while the most modern form was developed by [Carlson](#). Then there's the [Jacobi](#) forms, and there's also a set of forms called the [Weierstrass](#) functions. They can be confusing. I'm not going to do anything to reduce that confusion, other than to state that we'll be using the Jacobian version.

The link between elliptic integrals and the elliptic functions described above has a couple of steps. First we consider an incomplete Jacobian elliptic integral of the first kind.

$$u = F(\phi, k) = \int_0^\phi \frac{d\theta}{\sqrt{1 - k^2 \sin^2(\theta)}}$$

Here, ϕ , is called the *Jacobi Amplitude*, and it can be calculated using the *amplitude function*, $\text{am}(u, k)$. The amplitude function is the *inverse* of the incomplete elliptic function of the first kind $F(\phi, k)$. The amplitude function outputs a ϕ .

$$F^{-1}(u, k) = \text{am}(u, k) = \phi$$

Which we can then input into our standard trigonometric sine function, to

obtain our elliptic sine function $\text{sn}()$.

$$\text{sn}(u, k) = \sin(\text{am}(u, k)) = \sin(\phi)$$

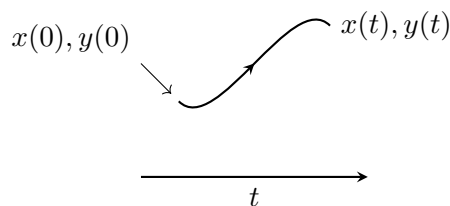
As an aside, you can define another important elliptic function, $\text{dn}(u)$ as the derivative of the amplitude function like so: $\text{dn}(u) = \frac{d}{du} \text{am}(u, k)$. Thus its name, the delta amplitude! This function can be also be formally defined in relationship to $\text{sn}(u)$: $\text{dn}(u) = \sqrt{1 - k^2 \text{sn}^2(u)}$.

Ok, we now have some tenuous understanding of a class of special functions called elliptic functions. We know that these functions are related to elliptic integrals, and we know that the integral we're trying to solve is an elliptic integral. We also know that any solution will involve some combination of our elliptic functions, F , E and Π and probably some elementary functions as well.

4 Arc Length Parametrization

We have to take a slight digression here to provide some more context. Without it we won't be able to make the conceptual leap the next section requires. This whole integral began with an attempt to find the arclength of the trefoil knot. What exactly is the arclength and why are we using it?

We are studying a special type of curve, a knot in $\mathbb{R}^3$. To use the tools of calculus on the knot, our functions have to have certain properties. One way to construct functions that have these properties is to use *parametrization*. We use a parameter t which represents time. It's helpful to think of a general curve function $\alpha(t)$, which is the path of a particle in space. As time advances the curve is traced out by the particle.



As we saw with our initial parametrization of the trefoil knot, the three coordinates, x , y and z each have their own parametrization function, all of which are controlled by the same parameter. Parametrizations are not unique, and we can describe a curve with multiple parametrizations.

Formally, a curve, α , maps an interval I on the real line $\mathbb{R}$ to some $\mathbb{R}^n$. Here's that statement in the tersest notation possible.

$$\alpha : I \rightarrow \mathbb{R}^n$$

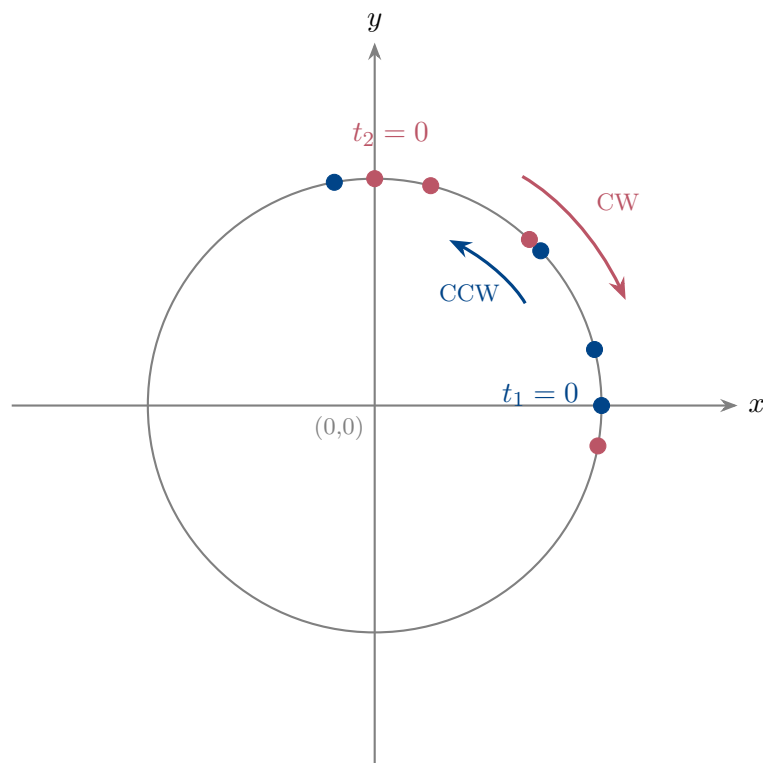
Then our parameter $t \in I$ is used to generate a value for each coordinate in the $\mathbb{R}^n$ our curve $\alpha(t)$ is mapping too. Parametrization lets us control the direction and weirdly, the speed with which a curve is traced out. Lets see how this happens with the unit circle. The standard parametrization of the unit circle matches the one we all learn in grade school.

$$\begin{aligned}\alpha_1(t) &= (x_1(t), y_1(t)) \\ x_1 &= \cos(t) \\ y_1 &= \sin(t)\end{aligned}$$

But we could just as easily define the opposite parametrization.

$$\begin{aligned}\alpha_2(t) &= (x_2(t), y_2(t)) \\ x_2 &= \sin(t) \\ y_2 &= \cos(t)\end{aligned}$$

How do these two parametrizations behave? Let's take some points on our interval $I = [0, 2\pi)$ and see. We'll calculate the position of x and y when $t = 0, 0.25, 0.75, 1.75$. Here's a diagram showing the difference.



Parametrization 1 in blue starts at the positive x -axis and continues in a counter-clockwise direction while parametrization 2 in red starts at the positive y -axis and continues in a clockwise direction. If we take the derivative of each of these parametrizations, we obtain an element called the *velocity vector*. Let's look at the derivative function $\alpha'_n(t)$.

$$\alpha'_n(t) = (x'_n(t), y'_n(t))$$

No es complicado. So for our first parametrization, $\alpha_1(t)$, taking the derivative gives us the following functions.

$$\begin{aligned} x'_1 &= -\sin(t) \\ y'_1 &= \cos(t) \end{aligned}$$

Inputting a value for t , would gives us a tangent vector to our curve giving its velocity and direction for that particular value of t . I mentioned earlier that we could control the speed of our parametrization. Let's see how that works. Again we can define a parametrization for our circle, we'll call it $\alpha_3(t)$.

$$\begin{aligned} x_3 &= \cos\left(\frac{t}{4}\right) \\ y_3 &= \sin\left(\frac{t}{4}\right) \end{aligned}$$

And if we find the velocity vector $\alpha'_3(t)$ we obtain the following.

$$\begin{aligned} x'_3 &= -\frac{\sin(\frac{t}{4})}{4} \\ y'_3 &= \frac{\cos(\frac{t}{4})}{4} \end{aligned}$$

So our new parametrization is 4 times slower than our initial parametrization. I've skipped the calculation but the speed of $\alpha'(t)$ is it's *norm*, $|\alpha'(t)| = \sqrt{x'(t)^2 + y'(t)^2}$. You can run the numbers yourself if you'd like. There are lots of other details to this stuff, you can read up about differential geometry if you want to know more(I do!). For now however I think we've got enough context to get to the next step.

5 Invert, Always Invert

The famous quote above about how to solve math problems comes appropriately from Carl Jacobi, after whom the Jacobian elliptic functions are

named. Here we'll discuss Jacobi's method of inversion and how its used to solve an elliptic integral.

We want to know how Jacobi found the special elliptic functions for the first (F), second (E) and third (Π) kind. Elliptic integrals can't actually be solved with elementary functions, so how did he discover their solutions? Jacobi's insight, was to look at the inverse of the integral, rather than the integral itself.

It's illuminating to consider an analogous situation with a circle. This discussion comes directly from John Stillwell's excellent Mathematics and Its History [14]. Consider $\sin(u)$, it takes in an argument u and outputs a number x .

$$\sin(u) = x$$

The *inverse* of the sine function takes our output x and returns our input u .

$$\sin^{-1}(x) = u$$

Sometimes its also called the $\arcsin()$ function. Moving to some algebra and recalling Pythagoras, lets also state another important relation about the unit circle.

$$x^2 + y^2 = 1$$

An easy bit of algebra will show that $y = \sqrt{1 - x^2}$, and thus by extension $\frac{1}{y} = \frac{1}{\sqrt{1 - x^2}}$. We're going to collect just a few more facts before we draw this whole thing together. Interestingly, the inverse sine function $\sin^{-1}(x)$ has an integral associated with it.

$$\sin^{-1}(x) = \int_0^x \frac{dt}{\sqrt{1 - t^2}}$$

The t in the integral is just a place holder variable, but the similarity with our value for $\frac{1}{y}$ should be quite apparent. We've now got the ability to parametrize our functions for x and y with the same variable, u . Let's see how that works out. First, for some notational ease, I'm going let $x = f(u) = \sin(u)$, so by definition $f'(u) = \frac{dx}{du}$ which is equal to $1/\frac{du}{dx}$. That might appear to be a bit of notational trickery (of course it is), but it's useful! Let's put these most recent facts together in a chain.

$$\frac{du}{dx} = \frac{d}{dx} \int_0^x \frac{dt}{\sqrt{1 - t^2}} = \frac{1}{\sqrt{1 - x^2}} = \frac{1}{y}$$

Thus $y = f'(u)$. And recalling our elementary calculus, we know that the derivative of $\sin$ is $\cos$ so $y = \cos(u)$ as well! Interestingly this specific case

generalizes, which allows us to draw an analogy with the elliptic integral of the first kind. Here it is all nice and pretty.

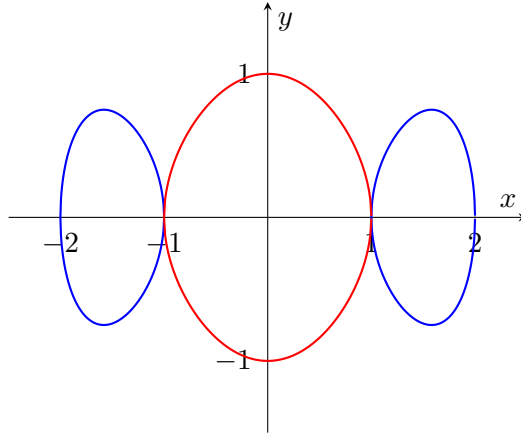
$$F(\phi, k) = \int_0^\phi \frac{d\theta}{\sqrt{1 - k^2 \sin^2(\theta)}}$$

There is also a coordinate form that doesn't use ϕ and instead uses x as it's integration variable.

$$\text{am}^{-1}(x, k) = F(\sin^{-1}(x), k) = \int_0^x \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}$$

So, this form is slightly different from the form that uses $\sin^2$ in its denominator. We can swap between it and that form using a u -substitution where set $x = \sin(\theta)$. We'll use this form right now because it compares well with the circle example we just saw.

We can see the obvious similarity in the structure of the integral to our circle example. It is important to recall the basic definitions of calculus, and understand that an integral and a derivative are inverse functions. Niels Henrik Abel was the first mathematician to publish the idea that an elliptic integral could be studied via its *inverse* (Gauss had worked it out earlier in his notes but had not published). Jacobi read Abel's work, and built upon it, giving us the Jacobian elliptic functions we know today. Historically the integral was all these mathematical greats had, and their insight was to *define* a corresponding function as its inverse (John Cook has a neat [post\[5\]](#) about this). Now lets derive the Elliptic Integral of the First Kind! To do that, I have to introduce a special curve called the *quartic*.



The formula for this specific quartic curve is as follows: $y^2 = (1 - x^2)(1 - 0.25x^2)$, and the general form is $y^2 = (1 - x^2)(1 - k^2x^2)$. The blue curves represent the imaginary components, while red signifies real values. We've set $k^2 = 0.25$ for the diagram. I should note that historically, this quartic

curve appeared when mathematicians were trying to calculate the perimeter of the ellipse. It didn't appear in the manner I'm introducing it though, I've just done that because I think it flows well with the earlier inverse analogy involving the circle. We can see how to obtain y very easily.

$$y = \sqrt{(1-x^2)(1-k^2x^2)}$$

So as before, we set the inverse of y , $\frac{1}{y} = \frac{1}{\sqrt{(1-x^2)(1-k^2x^2)}}$. Let's return to our Elliptic Integral of the First Kind. Recall that x is the upper limit of integration for this integral. As well we should remember that $x = \text{am}(u)$ for a fixed k . We can then define the inverse function.

$$u = \text{am}^{-1}(x) =$$

Well informed readers will know that since the amplitude function is the inverse of the elliptic sine function $\text{am}^{-1}(x) = \text{sn}(x)$. Thus, for some fixed k we have the following relationship.

$$\text{am}^{-1}(x) = \int_0^x \frac{d\theta}{\sqrt{(1-x^2)(1-k^2x^2)}}$$

We need to show that the link between y and our integral now. So as before we let $x = f(u) = \text{am}(u)$. Then $f'(u) = \frac{dx}{du} = 1/\frac{du}{dx}$, and we have the following chain of statements.

$$\frac{du}{dx} = \frac{d(\text{am}^{-1}(x))}{dx} = \frac{d}{dx} \int_0^x \frac{d\theta}{\sqrt{(1-x^2)(1-k^2x^2)}}$$

As before the integral and derivative cancel and that leaves us with the following.

$$\frac{du}{dx} = \frac{1}{\sqrt{(1-x^2)(1-k^2x^2)}} = \frac{1}{y}$$

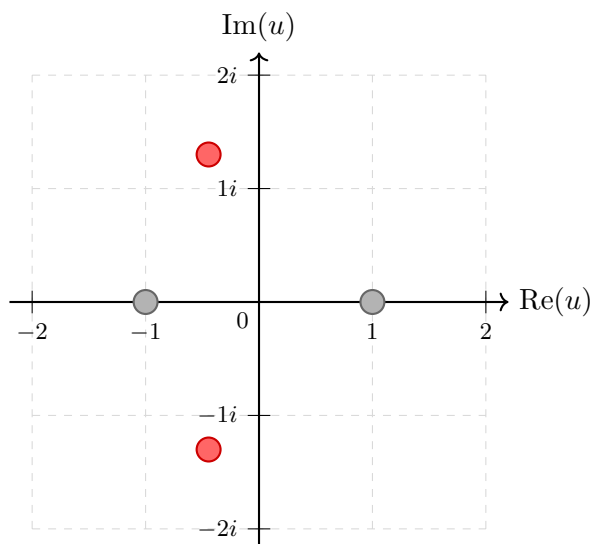
Thus, as before $y = f'(u)$ since $f'(u) = \frac{dx}{du}$ the inverse of $\frac{1}{y}$. I think, unfortunately, that when you walk through the steps of this technique there doesn't seem to be a grand revelation. It just reads as a series of true statements. The real genius here is that this method lets us find a solution to an integral that we can't solve with elementary functions. I like to think of the solution as being almost *invented*. Some of the greats had the audacity to just *name* a solution and then show the link afterwards. I hope to one day have that kind of confidence.

6 Complex Roots

The next step with a problem like this is to look at some solved elliptic integrals and hope that your problem either matches one, or is close to one.

Unfortunately after looking at a couple of key sources, like the Handbook of Elliptic Integrals for Engineers and Scientists[3], and the Table of Integrals, Series and Products by Gradshteyn and Ryzhik [9] nothing seemed to match. To use these tables we have to transform our integral into a different form. To do so, we're going to use a tool from complex analysis, the möbius transform (or linear fractional transform).

To begin lets recall the fundamental theorem of algebra, that a polynomial of degree n has exactly n complex roots. We want to look at the denominator in our integral (8), where we see two polynomials $(1 - u^2)$ and $(9u^2 + 8u + 17)$. The roots for the first are easy to determine, ± 1 . You can use the quadratic equation to find the roots of the second and you get the following ugly pair $u = -\frac{4}{9} \pm i\frac{\sqrt{137}}{9}$. We'll plot all these roots so we can see what's going on geometrically.



We want the transform to align our complex roots with the imaginary axis of our chart above. The usual quadratic has form $ax^2 + bx + c$. A polynomial equation of the form $ax^2 + c$ will have its two roots aligned to either the imaginary or real lines. This means we need to remove the $8u$ term (our bx) in our second polynomial. It's instructive to quickly work through the algebra to see why. Let's solve for x .

$$x = \pm \sqrt{-\frac{c}{a}}$$

If the term under the square root is negative, then we'll have a pair of imaginary roots, and a pair of real roots if its not. This form is handy since the elliptic integral tables assume a denominator polynomial function with its roots on either the complex or real axes.

7 Möbius Transforms

A möbius transform, or linear fractional transform, or bilinear transform (yes there are three names) has the following general form.

$$f(z) = \frac{az + b}{cz + d}$$

It's one example of a technique known as *conformal* (angle-preserving) *mapping*. Intuitively, a conformal map stretches or warps the complex plane $\mathbb{C}$ like it's a sheet of rubber. We want this stretching or warping effect because we want to keep two of our poles in the same place ± 1 , while we want to map the other two to the imaginary axis. So we already know one characteristic of our möbius transform, it has to keep a pole at ± 1 in the same spot. Let's introduce a möbius transform that does just that.

$$u = \frac{1 + Az}{A + z}$$

We should see that from our general form, using the following substitutions, $a = d = A$ and $c = b = 1$, generates this transform. If $z = \pm 1$ our $u = \pm 1$. Let $z = 1$ then:

$$u = \frac{1 + A}{A + 1} = 1$$

And if we let $z = -1$ then:

$$u = \frac{1 - A}{A - 1} = -1$$

So the first goal of the transform is achieved, maintaining our roots at ± 1 for any value of A . Of course now we need to find A ! In particular we want an A that gives our other pair of roots a form $z = 0 + ib$. As a first step we'll take our polynomial with complex roots and just substitute in our transform for u . That gives us the following equation.

$$9\left(\frac{1 + Az}{A + z}\right)^2 + 8\left(\frac{1 + Az}{A + z}\right) + 17 = 0$$

We want to simplify some things so if we multiply every element by $(A + z)^2$ we obtain the following.

$$9(1 + Az)^2 + 8(1 + Az)(A + z) + 17(A + z)^2 = 0$$

We can foil everything out and gather up our terms, but our goal here is to find an A such that our linear term $8u$ disappears. So we're going to gather our linear terms (terms with a single z after substitution) and find out what value of A makes them 0. Rather than be overwhelmed by a whack of algebra we'll consider each term independently.

$$9(1 + Az)^2 = 9(1 + 2Az + (Az)^2) = 9 + 18Az + 9(Az)^2$$

So our first linear component is $18Az$, here's our second.

$$8(1 + Az)(A + z) = 8(A + z + A^2z + Az^2) = 8A + 8(1 + A^2)z + 8Az^2$$

Then our final is as follows.

$$17(A + z)^2 = 17(A^2 + 2Az + z^2) = 17A^2 + 34Az + 17z^2$$

So we'll assemble these three components and solve for A . We can of course ignore z for now.

$$18A + 8(1 + A^2) + 34A = 8 + 52A + 8A^2 = 0$$

After some a quick division by 4 we obtain the following.

$$2 + 13A + 2A^2 = 0$$

Now we can deploy our trusty high school companion, the quadratic equation, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$A = \frac{-13 \pm \sqrt{13^2 - 4(2)^2}}{2(2)} = \frac{-13 \pm 3\sqrt{17}}{4}$$

Voila! We have a value for A that will remove the middle term in our polynomial!

We've deployed this möbius transform, but we haven't really shown enough to understand how to use it more generally or why it works. Here's a bit more context that helps explain that. First, we have to understand another factor called the cross-ratio. One of the revolutions that occurred in 19th Century mathematics was prompted by Felix Klein's Erlanger Program. This program studied geometry through the lens of invariance under transformation. Speaking generally, the idea was these things that were unchanged by a transformation, the invariants, were the important properties of a geometric space. So we have a transformation, the möbius transform, and the cross-ratio is its fundamental invariant! Formally the cross-ratio, λ , is defined using four points in the complex plane (z_1, z_2, z_3, z_4) .

$$\lambda = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)}$$

We'll pick 4 coordinates, $z_1 \dots z_4$ and input them into our möbius transform, this will give us 4 new coordinates $u_1 \dots u_4$. Then we just have to calculate two cross-ratios. So we'll have an initial cross-ratio λ_i (from our z_n 's) and a final cross-ratio λ_f (from our u_n 's) and we want to make sure

that $\lambda_i = \lambda_f$. If this condition holds, we haven't changed our underlying equation in a fundamental way. Technically our transformed polynomial is the same function, just viewed from a different perspective in the *projective plane*. There's not enough space to go into the projective plane here, but suffice to say it's a fascinating object that pops up in things like art, perspective and photography. In our case the möbius transform produces the same cross-ratio for our initial and transposed points. Let's check that now! Here are our 4 starting coordinates.

$$z_1 = i$$

$$z_2 = 1$$

$$z_3 = -1$$

$$z_4 = -i$$

Before we transform them, let's calculate our initial cross ratio λ_i

$$\lambda_i = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)} = \frac{(i + 1)(1 + i)}{(1 + 1)(i + i)} = \frac{1}{2}$$

Now we can generate our 4 transformed coordinates. We've already shown that our transform preserves ± 1 so I can just restate them.

$$u_2 = z_2 = 1$$

$$u_3 = z_3 = -1$$

Then we need to calculate our final two u coordinates.

$$u_1 = \frac{1 + Az_1}{A + z_1} = \frac{1 + Ai}{A + i}$$

$$u_4 = \frac{1 + Az_4}{A + z_4} = \frac{1 - Ai}{A - i}$$

And then we calculate our final cross-ratio λ_f .

$$\lambda_f = \frac{(u_1 - u_3)(u_2 - u_4)}{(u_2 - u_3)(u_1 - u_4)} = \frac{(u_1 + 1)(1 - u_4)}{(1 + 1)(u_1 - u_4)} = \frac{(u_1 - u_1u_4 + 1 - u_4)}{2(u_1 - u_4)}$$

Since that numerator is a little gnarly we'll break it out separately and simplify things. It should be apparent by examination, that if we can find $u_1u_4 = 1$ everything will simplify into $\frac{1}{2}$. Here's the numerator reordered to make it explicit.

$$1 - u_1u_4 + u_1 - u_4$$

Then we can look at $u_1 u_4$. We'll keep A instead of subbing in a numeric value for now.

$$u_1 u_4 = \frac{(1 + Ai)(1 - Ai)}{(A + i)(A - i)} = \frac{1 - Ai + Ai - A^2 i^2}{A^2 - Ai + Ai - i^2} = \frac{1 + A^2}{A^2 + 1} = 1$$

We can then swap in our value for $u_1 u_4$.

$$\lambda_f = \frac{(u_1 - 1 + 1 - u_4)}{2(u_1 - u_4)} = \frac{(u_1 - u_4)}{2(u_1 - u_4)} = \frac{1}{2}$$

As we expected $\lambda_i = \lambda_f$. It's important to understand that we're just validating the behaviour of the transform itself with this check. We've run a set of test data through it.

8 Shut up and calculate

Ok, we're in a position where we've got lots of context, lets do some ALGEBRA! First lets recall our integral, of course its stripped of that constant out front to keep things simple.

$$\int \frac{17 + 8u + 9u^2}{\sqrt{1 - u^2} \sqrt{17 + 8u + 9u^2}} du$$

We spent a bunch of time figuring out our möbius transform. Rather than clutter things up with numbers, we'll just remember that $A = \frac{-13 \pm 3\sqrt{17}}{4}$. Let's attack the integral with the transform. As always with a substitution we have to figure out how to transform the du term as well, not just the u . Recall our transform.

$$u = \frac{1 + Az}{A + z}$$

Then du is found as follows using the quotient rule.

$$du = \frac{A(A + z) - (1 + Az)}{(A + z)^2} dz = \frac{A^2 - 1}{(A + z)^2} dz$$

So lets proceed! Subbing in our transform produces the following monstrosity.

$$\int \frac{17 + 8(\frac{1+Az}{A+z}) + 9(\frac{1+Az}{A+z})^2}{\sqrt{1 - (\frac{1+Az}{A+z})^2} \sqrt{17 + 8(\frac{1+Az}{A+z}) + 9(\frac{1+Az}{A+z})^2}} \frac{(A^2 - 1)}{(A + z)^2} dz \quad (9)$$

That's a lot of letters scattered around, we need to simplify things. Let's take it the same way New Kids On The Block do, Step by Step. As an engineer I've been trained to do the easiest part first. So we'll attack the following piece.

$$1 - u^2 = 1 - \left(\frac{1 + Az}{A + z} \right)^2$$

The first step is to bring the 1 into the numerator and then foil everything out up top.

$$\frac{(A + z)^2 - (1 + Az)^2}{(A + z)^2} = \frac{A^2 + 2Az + z^2 - 1 - 2Az - A^2z^2}{(A + z)^2}$$

We can cancel like terms and rearrange and that gets us the following.

$$\frac{(A^2 - 1) - z^2(A^2 - 1)}{(A + z)^2} = \frac{(A^2 - 1)(1 - z^2)}{(A + z)^2}$$

Next we put it back under the square root and we're done the easiest and first part (note that $(A + z)$ isn't squared since it and the root cancel)!

$$\sqrt{1 - u^2} = \frac{\sqrt{(A^2 - 1)(1 - z^2)}}{(A + z)}$$

Just for a bit more organization in our integral we'll give the form under the square root a name.

$$P(z) = \frac{(A^2 - 1)(1 - z^2)}{(A + z)^2}$$

Ok, now we'll move on to the next term in the denominator.

$$9 \left(\frac{1 + Az}{A + z} \right)^2 + 8 \left(\frac{1 + Az}{A + z} \right) + 17$$

This one is a little bit gnarlier than before. Lets make it a little easier to read.

$$\frac{9(1 + Az)^2 + 8(1 + Az)(A + z) + 17(A + z)^2}{(A + z)^2}$$

Lets foil it all out again.

$$\frac{9(1 + 2Az + (Az)^2) + 8(A + z + A^2z + Az^2) + 17(A^2 + 2Az + z^2)}{(A + z)^2}$$

Now we'll group everything by powers of z .

$$\frac{9 + 8A + 17A^2 + 8z + 52Az + 8A^2z + 9A^2z^2 + 8Az^2 + 17z^2}{(A + z)^2}$$

Note the symmetry between the z^0 and z^2 terms (in red). The linear terms in z are in blue and have a different form. We should recognize that

form from our solution for A ! We can zero out those terms since we've already shown that that's what they equal. Next I'm going to name the z^0 and z^2 elements. Here's z^0 .

$$C_1 = 9 + 8A + 17A^2$$

And then our z^2 component.

$$C_2 = 9A^2 + 8A + 17$$

Now let's put together a function $Q(z)$.

$$Q(z) = \frac{C_1 + C_2 z^2}{(A + z)^2} = \frac{C_1(1 + \frac{C_2}{C_1} z^2)}{(A + z)^2}$$

Ok, so we've dealt with all of the substitutions in our integral (9) and we can return to it now.

$$\int \frac{Q(z)}{\sqrt{P(z)}\sqrt{Q(z)}} \frac{(A^2 - 1)}{(A + z)^2} dz \quad (10)$$

I think it mostly looks nicer because we've put a bunch of stuff inside of $Q(z)$ and $P(z)$. But, we can take heart that we have zeroed out some terms, and it looks like there are some opportunities for cancelling some other terms too (the equation has gone woke!). Deep breath now and let's attack the beast.

$$\int \frac{\frac{C_1(1 + \frac{C_2}{C_1} z^2)}{(A + z)^2}}{\frac{\sqrt{(A^2 - 1)(1 - z^2)}}{(A + z)} \sqrt{C_1(1 + \frac{C_2}{C_1} z^2)}} \frac{(A^2 - 1)}{(A + z)^2} dz \quad (11)$$

Ok, there's some immediate opportunities for simplification here. We can eliminate some of the $(A + z)$ terms. That gives us the following below. The step isn't that complicated so I've skipped writing it out by hand, but if you want a breakdown, feel free to ask a program like Maple or Wolfram or have an LLM walk you through it.

$$\int \frac{C_1(1 + \frac{C_2}{C_1} z^2)}{\sqrt{(A^2 - 1)(1 - z^2)}\sqrt{C_1(1 + \frac{C_2}{C_1} z^2)}} \frac{(A^2 - 1)}{(A + z)^2} dz \quad (12)$$

Then we can combine a few more elements to get a simpler equation.

$$\int \frac{\sqrt{C_1(1 + \frac{C_2}{C_1} z^2)}(A^2 - 1)}{\sqrt{1 - z^2}(A + z)^2} dz \quad (13)$$

Ok, that looks a little less intimidating, we just have to remember that C_1 and C_2 are dependent on our value of A and quadratic. The hawk-eyed

amongst you should also see a constant term we can pull out of the integral. $\sqrt{(A^2 - 1)}$. Let's combine that with our previous constant $-\frac{1}{3}$, we'll call the constant in front our integral G . Here's its current value.

$$G = -\frac{\sqrt{A^2 - 1}}{3}$$

So here's the outcome of all the work we've done with the constant term removed. The integral looks quite different, and we no longer have a quadratic with a linear term in either our numerator or denominator.

$$\int \frac{\sqrt{C_1 + C_2 z^2}}{\sqrt{1 - z^2}(A + z)^2} dz \quad (14)$$

All this work lets us access the elliptic integral tables that we looked at earlier.

9 Cometh the Hour

The first step of the next step is to revert some of our substitutions. Recall that $z = \cos(\phi)$ from equation (6). As always, we've got to make sure we handle the differential substitution correctly, here $dz = -\sin(\phi)d\phi$. That gives us the following.

$$\int \frac{\sqrt{C_1 + C_2 \cos^2(\phi)}}{\sqrt{1 - \cos^2(\phi)}(A + \cos(\phi))^2} (-\sin(\phi)) d\phi \quad (15)$$

As lovers of trigonometry, we should see that $\sin(\phi) = \sqrt{1 - \cos^2(\phi)}$ which of course cancels out the sin that popped in with our differential substitution. That's very nice!

$$\int \frac{\sqrt{C_1 + C_2 \cos^2(\phi)}}{(A + \cos(\phi))^2} d\phi \quad (16)$$

And we pull the negative out to modify G again.

$$G = \frac{\sqrt{A^2 - 1}}{3}$$

Ok, the next step is to modify our denominator to look a bit more like $\text{dn}(u)$ which we should recall as the delta term, $\text{dn}(u) = \Delta(\phi) = \sqrt{1 - m \sin^2(\phi)}$. We were briefly introduced to this elliptic function earlier. Both the elliptic integrals of the first and second kind include this delta term. Lets look at the numerator of our integral.

$$\sqrt{C_1 + C_2 \cos^2(\phi)}$$

We'll perform our trusty trig substitution where $\cos^2(\phi) = 1 - \sin^2(\phi)$.

$$\sqrt{C_1 + C_2(1 - \sin^2(\phi))} = \sqrt{(C_1 + C_2) - C_2 \sin^2(\phi)}$$

Then we can factor out our C_1 and C_2 terms like so.

$$\sqrt{C_1 + C_2} \sqrt{1 - \frac{C_2}{C_1 + C_2} \sin^2(\phi)}$$

We can quickly see that $m = \frac{C_2}{C_1 + C_2}$, and after we remove our constant term our integral looks like this.

$$\int \frac{\sqrt{1 - \frac{C_2}{C_1 + C_2} \sin^2(\phi)}}{(A + \cos(\phi))^2} d\phi = \int \frac{\Delta(\phi)}{(A + \cos(\phi))^2} d\phi \quad (17)$$

Let's not forget our G out front, here's what it's now equal to.

$$G = \frac{\sqrt{(A^2 - 1)(C_1 + C_2)}}{3}$$

There's a few more steps to take. If we peek at our tables of elliptic integrals, we have something that is close to some forms, but doesn't match exactly. Lets do a couple more changes to our integral.

$$\int \frac{\Delta(\phi)}{(A + \cos(\phi))^2} d\phi = \int \frac{\Delta^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

Now our numerator doesn't have a square root in it anymore and we can split the integral.

$$\int \frac{1 - m \sin^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi = \int \frac{d\phi}{(A + \cos(\phi))^2 \Delta(\phi)} - \int \frac{m \sin^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

To make tracking these integrals a bit easier I'm going to call the first I_1 and the second I_2 .

$$I_1 = \int \frac{d\phi}{(A + \cos(\phi))^2 \Delta(\phi)}$$

$$I_2 = \int \frac{m \sin^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

So our complete formula looks like this at the moment. Nice and simple!

$$G(I_1 - I_2)$$

9.1 The First Half

That first integral, I_1 is known and solved! We'll look at Byrd and Friedman, pg. 205, integral 341. Here's the integral form they provide.

$$R_n = \int \frac{d\phi}{(1 + \alpha \cos(\phi))^n \sqrt{1 - k^2 \sin^2(\phi)}}$$

There are some notational differences, they're using k^2 instead of m , and we have to factor out A (like so, $A(1 + \frac{1}{A} \cos(\phi))$), but nothing major. Let's setup our specific case, we'll swap the term in front of $\sin(\phi)$ so it matches our notation and we'll let $\alpha = \frac{1}{A}$. Then we have the following.

$$R_2 = \int \frac{d\phi}{(1 + \alpha \cos(\phi))^2 \sqrt{1 - m \sin^2(\phi)}}$$

We're not going to update G for this integral, but we should track the individual terms that we can pull out of the integral. Here, the A^2 term comes out after you pull the factored A from the squared brackets in the denominator of the integral. Since $\alpha = \frac{1}{A}$, $\frac{1}{A^2} = \alpha^2$. So we have $\alpha^2 R_2$ at this stage. Returning to the integral R_2 , there are some conditions we have to check when using these tables. Byrd and Friedman solve this specific case with 341.04. Here it is written out, please note I've swapped k 's to match our notation. Recall that $k^2 = m$ and $k'^2 = 1 - k^2 = 1 - m$.

$$R_2 = \frac{1}{(\alpha^2 - 1)(m + \alpha^2(1 - m))} \left\{ (\alpha^2(2m - 1) - 2m)R_1 + 2mR_{-1} - mR_{-2} + \frac{\alpha^2 \operatorname{sn}(u) \operatorname{dn}(u)}{1 + \alpha \operatorname{cn}(u)} \right\}$$

The various R_n terms scattered throughout, correspond to functions that are provided by Byrd and Friedman. Here's the first.

$$R_1 = \frac{1}{1 - \alpha^2} \left[\Pi(\phi, \frac{\alpha^2}{\alpha^2 - 1}, \sqrt{m}) - \alpha f_1 \right]$$

Here Π is an elliptic integral of the third kind and f_1 is another function defined later by Byrd and Friedman. The form of f_1 depends on the values of α and m . In our case since $\frac{\alpha^2}{\alpha^2 - 1} < m$, f_1 has the following form.

$$f_1 = \sqrt{\frac{1 - \alpha^2}{m + (1 - m)\alpha^2}} \tan^{-1} \left[\sqrt{\frac{m + (1 - m)\alpha^2}{1 - \alpha^2}} \operatorname{sd}(u) \right]$$

Note that $\operatorname{sd}(u)$ is an elliptic function. The next function is R_{-1} which is mercifully short.

$$R_{-1} = u + \frac{\alpha}{\sqrt{m}} \cos^{-1}(\operatorname{dn}(u))$$

And finally we have R_{-2} .

$$R_{-2} = \frac{1}{m}[(m - \alpha^2(1 - m))u + \alpha^2 E(u) + 2\alpha\sqrt{m} \cos^{-1}(\text{dn}(u))]$$

Here E is an elliptic integral of the second kind. We're going to run screaming from this whole thing right now and turn to the second integral. Hopefully some magical cancellations will appear and let us write it in a more compact form.

9.2 The Second Half

Ok we've got one more integral to go. Let's recall it.

$$I_2 = \int \frac{m \sin^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

We've got a constant term for this integral, we can pull out m , giving us the following.

$$m \int \frac{\sin^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

As always we'll use our by now favourite trigonometric identity and swap $\sin^2(\phi)$ for $1 - \cos^2(\phi)$.

$$m \int \frac{1 - \cos^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

We need to transform our numerator somehow, our substitution gives us a different form but doesn't change anything fundamental. However, let's recall the difference of squares relation, that $a^2 - b^2 = (a - b)(a + b)$. We've got a $\cos(\phi)$ term in both our numerator and denominator. Maybe we can find a cancellation here? Lets perform that wonderful mathematical trick of adding nothing to our equation. We'll add $+A^2$ and $-A^2$ to our numerator.

$$m \int \frac{1 - A^2 + A^2 - \cos^2(\phi)}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

Then we perform some quick algebra and obtain the following form.

$$m \int \frac{1 - A^2 + (A - \cos(\phi))(A + \cos(\phi))}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

The cancellation is so close we can taste it! We just have to split our integral first.

$$I_2 = m \int \frac{1 - A^2}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi + m \int \frac{(A - \cos(\phi))(A + \cos(\phi))}{(A + \cos(\phi))^2 \Delta(\phi)} d\phi$$

We can cancel our term now! Hopefully we can also see that the integral on the left is exactly the same form as I_1 . The constant will be slightly different but that's a-ok. Just for notational consistency I'll name the first I_{21} and the second I_{22} . Here's the I_{21} with its form made explicit.

$$I_{21} = m(1 - A^2) \int \frac{d\phi}{(A + \cos(\phi))^2 \Delta(\phi)}$$

Let's return to I_{22} and take the just reward of our recent work (our cancellation).

$$I_{22} = m \int \frac{(A - \cos(\phi))}{(A + \cos(\phi)) \Delta(\phi)} d\phi$$

So we have a form that's meaningfully different now. Let's look at that fraction without the $\Delta(\phi)$ function and see if we can come up with something. We'll reuse that trick of adding nothing like so.

$$\frac{A + A - A - \cos(\phi)}{A + \cos(\phi)} = \frac{2A - (A + \cos(\phi))}{A + \cos(\phi)} = \frac{2A}{A + \cos(\phi)} - 1$$

If we put this back into our integral we get the following.

$$I_{22} = m \int \frac{2A}{(A + \cos(\phi)) \Delta(\phi)} d\phi - m \int \frac{d\phi}{\Delta(\phi)}$$

That second integral looks pretty simple and is just an elliptic integral of the first kind, F . Here's the actual definition below, it should be pretty apparent how similar things are.

$$F(\phi, k) = \int_0^\phi \frac{d\theta}{\Delta(\theta)}$$

That was easy! The first integral is also solved and is actually just the R_1 case we looked at when using Byrd and Friedman for the first half of the integral. Here's the specific case

$$R_1 = \int \frac{d\phi}{(1 + \alpha \cos(\phi)) \sqrt{1 - m \sin^2(\phi)}}$$

We can of course pull out that constant on top so we have $2mR_1$ (the A cancels when we factor it out of the denominator). And then rather than looking back a few pages, here's R_1 restated.

$$R_1 = \frac{1}{1 - \alpha^2} \left[\Pi(\phi, \frac{\alpha^2}{\alpha^2 - 1}, \sqrt{m}) - \alpha f_1 \right]$$

9.3 Putting it Back Together

Ok! So we have everything solved, we just have to put it together and maybe find some simplifications. It's funny, at this level of mathematics, naming some of these functions reminds me of how I used to clean my room for my parents when I was a teenager. Just stuff it all in the closet. Here G is our constant and I_n are our integrals.

$$G(I_1 - I_2) = G(I_1 - I_{21} - I_{22})$$

Those are the three integrals we solved for and here are their named solutions (I've suppressed the function inputs for F to keep the equation shorter).

$$G(I_1 - I_{21} - I_{22}) = G(\alpha^2 R_2 - m(1 - A^2)\alpha^2 R_2 - 2mR_1 + mF)$$

So we can do a little bit of algebra here given the two R_2 's we've got.

$$\alpha^2 R_2 - m(1 - A^2)\alpha^2 R_2 = R_2(\alpha^2 - m(1 - A^2)\alpha^2)$$

We'll just look at the constant term now, and we should recall that $\alpha^2 = \frac{1}{A^2}$.

$$\alpha^2 - m(1 - A^2)\alpha^2 = \alpha^2 - m(\alpha^2 - 1) = \alpha^2(1 - m) + m$$

This is a nice little term to have because it matches part of the denominator in front of R_2 . Here's what we have right now (we'll just consider the constants up front).

$$(\alpha^2(1 - m) + m)R_2 = (\alpha^2(1 - m) + m) \frac{1}{(\alpha^2 - 1)(\alpha^2(1 - m) + m)} \{...\}$$

Of course the fraction out front simplifies nicely, we'll call this version R_{2*} .

$$R_{2*} = \frac{1}{(\alpha^2 - 1)} \{...\}$$

Ok, let's return to the main equation and sub in our simplification.

$$G(I_1 - I_{21} - I_{22}) = G(R_{2*} - 2mR_1 + mF)$$

The next step is to unpack R_{2*} , it was a big gnarly integral.

$$R_{2*} = \frac{1}{(\alpha^2 - 1)} \left\{ (\alpha^2(2m - 1) - 2m)R_1 + 2mR_{-1} - mR_{-2} + \frac{\alpha^2 \operatorname{sn}(u) \operatorname{dn}(u)}{1 + \alpha \operatorname{cn}(u)} \right\}$$

Just to make it a little easier to write out, we'll let $L = \frac{\alpha^2 \operatorname{sn}(u) \operatorname{dn}(u)}{1 + \alpha \operatorname{cn}(u)}$. So that gives us the following.

$$R_{2*} = \frac{1}{(\alpha^2 - 1)} \{(\alpha^2(2m - 1) - 2m)R_1 + 2mR_{-1} - mR_{-2} + L\}$$

If we expand this in our complete integral expression we've got the full meal deal below.

$$G\left(\frac{1}{(\alpha^2 - 1)} \{(\alpha^2(2m - 1) - 2m)R_1 + 2mR_{-1} - mR_{-2} + L\} - 2mR_1 + mF\right)$$

We can see we've got a couple R_1 s so lets pull those out and look at them. We'll have the following once we foil out our fractional constant in front of R_{2*} .

$$\frac{(\alpha^2(2m - 1) - 2m)R_1}{(\alpha^2 - 1)} - 2mR_1 = \frac{(\alpha^2(2m - 1) - 2m)R_1 - 2mR_1(\alpha^2 - 1)}{(\alpha^2 - 1)}$$

We can distribute out everything in the denominator, which gives us the following expression.

$$\frac{(2mR_1\alpha^2 - R_1\alpha^2 - 2mR_1 - 2mR_1\alpha^2 + 2mR_1)}{(\alpha^2 - 1)}$$

Dealing with the obvious cancellations leaves us with the following.

$$-\frac{\alpha^2}{(\alpha^2 - 1)}R_1$$

Interestingly this is the same input we had for elliptic integral of the third kind in R_1 , in the general form that input is referred to as the *characteristic* and is given the symbol n . I'm not going to bother with that here though other than to note it! Let's get back to our equation and see what the general form is.

$$G\left(-\frac{\alpha^2}{(\alpha^2 - 1)}R_1 + \frac{2m}{\alpha^2 - 1}R_{-1} - \frac{m}{\alpha^2 - 1}R_{-2} + mF + \frac{L}{\alpha^2 - 1}\right) \quad (18)$$

Phew! Ok, that's all she wrote, so long as I haven't made a transcription error (I checked so many times) this is the final equation! Now we want something that we can calculate with. Let's sub in all our values and see what we get.

$$\begin{aligned}
& \frac{\sqrt{(A^2 - 1)(C_1 + C_2)}}{3} \left(-\frac{\alpha^2}{(\alpha^2 - 1)} \left[\frac{1}{1 - \alpha^2} \left(\Pi \left(\phi, \frac{\alpha^2}{\alpha^2 - 1}, \sqrt{m} \right) - \alpha f_1 \right) \right] \right. \\
& \quad + \frac{2m}{\alpha^2 - 1} \left(u + \frac{\alpha}{\sqrt{m}} \cos^{-1}(\text{dn}(u)) \right) \\
& \quad - \frac{1}{\alpha^2 - 1} [(m - \alpha^2(1 - m))u + \alpha^2 E(u) + 2\alpha\sqrt{m} \cos^{-1}(\text{dn}(u))] \\
& \quad + mF(\phi, \sqrt{m}) \\
& \quad \left. + \frac{\alpha^2 \text{sn}(u) \text{dn}(u)}{(\alpha^2 - 1)(1 + \alpha \text{cn}(u))} \right)
\end{aligned}$$

It's big, each line holds the constant and a single term (eg: R_1 , R_{-2} etc.) to make it a little more manageable to read. In theory you could compute this by hand but in practice you'll use a computer language like python or a program like Maple or Wolfram.

I should also note that this equation doesn't match Parcly Taxel's posted solution *exactly*. One key difference is that I use elliptic functions here, while Parcly's uses only standard trigonometric ones. I'm not going to prove their equivalence, this document is already much longer than I anticipated, and at some point in the future I'm going to write another where I use this formula for computation (like a good engineer!). I plan to show their equivalence then. I've made a very good effort to work through this math, and I believe it to be correct. If you need greater guarantees of accuracy, well, you can't just trust random mathematics you've downloaded off the internet can you? You'll have to put in a bit of work yourself.

10 My Vast Domain

I've been a little sloppy with the math here in that I haven't discussed the domain over which we're integrating. If we go all the way back to the beginning we start with our variable t . Since the trefoil knot is defined for $[0, 2\phi)$, that's our original domain for t . Our first substitution sets $\phi = 3t$ which gives us a domain for ϕ from $[0, 6\phi)$. The trefoil has a property called *three-fold symmetry*. This means we can break the knot into 6 identical 'half-lobes'. A 'half-lobe' of the trefoil knot will be traced out as t varies from 0 to $\frac{\pi}{3}$.

A *complete* elliptic integrals is generally defined over a set range from 0 to $\frac{\pi}{2}$. In practice what this means is that we will integrate a complete elliptic integral and multiply it by the periodicity of our input parameter ϕ to get the full result we want. So if we trace out a half lobe we'll multiply our result by 6 to get our full value for our complete trefoil. Let's recall the following trigonometric identity.

$$\cos(3t) = 1 - 2\sin^2\left(\frac{3t}{2}\right)$$

What happens when $t = 0$? In our sin we have $\frac{3(0)}{2} = 0$. When $t = \frac{\pi}{3}$ we have, $\frac{3(\frac{\pi}{3})}{2} = \frac{\pi}{2}$. Conveniently the input varies across the range of the complete elliptic integral. I plan to go into more detail about this when I actually compute with the integral this document is about. But for now that's enough!

11 Compression

Having done all of this work, what's the rule we should take from this? I don't know about you'd but I'd prefer not to have to go through this rigamarole again. Let's try and *generalize* just a little bit. We started with an integral of the following form.

$$\int \sqrt{a + b\cos(nt) + c\cos^2(nt)} dt$$

We can use our u -substitution to obtain a quadratic of the form $a + bu + cu^2$. The next goal here is to symmetrize the roots. To do that we need to find A for our transform $\frac{1+Au}{A+u}$. We do that by solving for A in the equation below.

$$bA^2 + 2(a + c)A + b = 0$$

Why this specific quadratic equation? Here's a short proof. If we plug our möbius transform into our quadratic $a + bu + cu^2$ we obtain the following equation (after removing the denominator).

$$a(A + u)^2 + b(1 + Au)(A + u) + c(1 + Au)^2 = 0$$

We can then go ahead and factor everything out.

$$a(A^2 + 2Au + u^2) + b(A + u + A^2u + Au^2) + c(1 + 2Au + A^2u^2) = 0$$

We can then collect the linear terms (highlighted in blue above), and we get the following group.

$$2Aau + bu + bA^2u + 2Acu$$

We do a bit of algebra on them, and voila!

$$u(bA^2 + 2A(a + c) + b)$$

We have the equation we started with! Since a , b and c will be given for any specific case, you just solve for the A that sets the equation to 0 and all your linear terms in u disappear! Then one will always have a quadratic equation of the form $C_1 + C_2 u^2$ and the constants C_1 and C_2 can be calculated using the following formulae.

$$C_1 = c + bA + aA^2$$

$$C_2 = cA^2 + bA + a$$

Then the last bit here is our value for m . Recall that $m = k^2$ and is the constant in front of $\sin$ in the denominator of the elliptic integral of the first kind.

$$m = \frac{C_2}{C_1 + C_2}$$

12 Conclusion

That's it. It's over, finished, kaput.

A Constant Calculations

This appendix provides a reference for values used in calculations throughout this document.

A.1 A

$$A = \frac{-13 \pm 3\sqrt{17}}{4}$$

There are two possible numerical values for A , since it has two roots. Both are approximated here to 4 significant digits.

$$A_+ = \frac{-13 + 3\sqrt{17}}{4} \approx -0.1577$$

$$A_- = \frac{-13 - 3\sqrt{17}}{4} \approx -6.342$$

Given these two values for A , it is natural to ask which to use going forward. We will use the value for A that comes from the negative square root, -6.342 . Using this value avoids a *singularity* in our calculations. Since we routinely run into $A + \cos(\phi)$ in our calculations, we want to avoid setting that term to 0. If it occurs in the denominator of a fraction, we have a singularity. Since $\cos(\phi)$ varies between -1 and 1 over the full period of ϕ , at some point it will obtain the value $+0.1577$. If we are using A_+ , our singularity then appears, and we cannot complete our calculation.

A.2 α

$$\alpha = \frac{1}{A_-} = \frac{4}{-13 - 3\sqrt{17}} \approx -0.1577$$

Interestingly the value for α appears to be the same as the positive root, A_+ . They are in fact exactly the same because the quadratic we solved is a *reciprocal equation*. This means that the two roots are reciprocals of one another. Thus $\frac{1}{A_+} = A_-$ and vice versa. Which is a neat little fact! I'm just going to state it and not prove it, but you can look up Vieta's formulas (formulae?) if you'd like to take that step.

A.3 C_1

Next up we have the constant values from our quadratic equation.

$$C_1 = 9 + 8A_- + 17A_-^2 = 9 + 8 \left(\frac{-13 - 3\sqrt{17}}{4} \right) + 17 \left(\frac{-13 - 3\sqrt{17}}{4} \right)^2$$

$$C_1 \approx 642.0$$

A.4 C_2

$$C_2 = 9A_-^2 + 8A_- + 17 = 9 \left(\frac{-13 - 3\sqrt{17}}{4} \right)^2 + 8 \left(\frac{-13 - 3\sqrt{17}}{4} \right) + 17$$

$$C_2 \approx 328.3$$

A.5 m

Finally we get to m which is a bit of a messy calculation.

$$\begin{aligned} m &= \frac{C_2}{C_1 + C_2} = \frac{9A_-^2 + 8A_- + 17}{26A_-^2 + 16A_- + 26} \\ &= \frac{9 \left(\frac{-13 - 3\sqrt{17}}{4} \right)^2 + 8 \left(\frac{-13 - 3\sqrt{17}}{4} \right) + 17}{26 \left(\frac{-13 - 3\sqrt{17}}{4} \right)^2 + 16 \left(\frac{-13 - 3\sqrt{17}}{4} \right) + 26} \end{aligned}$$

$$m \approx 0.3383$$

A.6 $1 - m$

$1 - m$ has a nice algebraic simplification we can do.

$$\begin{aligned}
 1 - m &= 1 - \frac{C_2}{C_1 + C_2} = \frac{C_1 + C_2 - C_2}{C_1 + C_2} = \frac{C_1}{C_1 + C_2} \\
 &= \frac{9 + 8A_- + 17A_-^2}{26A_-^2 + 16A_- + 26} \\
 &= \frac{9 + 8\left(\frac{-13-3\sqrt{17}}{4}\right) + 17\left(\frac{-13-3\sqrt{17}}{4}\right)^2}{26\left(\frac{-13-3\sqrt{17}}{4}\right)^2 + 16\left(\frac{-13-3\sqrt{17}}{4}\right) + 26} \\
 1 - m &\approx 0.6617
 \end{aligned}$$

A.7 Byrd and Friedman Conversion Table

Readers who would like to quickly check the conversion between Byrd and Friedman's notation and that used in this paper can reference the table here.

Concept	Byrd & Friedman	This Paper
Squared Modulus	k^2	m
Complementary Modulus	k'^2	$1 - m$
Amplitude	ϕ	ϕ
Delta Amplitude	$\sqrt{1 - k^2 \sin^2(\phi)}$	$\Delta(\phi)$
Modular Angle	α	$\frac{1}{A}$

B Math.StackExchange Question

I had asked a [question](#) that had already been asked by [Christian Fieldhouse](#) and Parcly Taxel was able to apply a [method](#) he developed for a previous integral question to it. Christian's question [\[6\]](#) is reproduced below in case the link provided earlier does not work..

Mathematics StackExchange: Arclength Parameterization of the Trefoil Knot

Asked by: Christian Fieldhouse **Date:** December 16, 2017
URL: <https://math.stackexchange.com/q/2569180>

Question:

I would like to find an arclength parameterization of the trefoil knot.
The parameterizations I can find are:

$$(\sin(t) + 2 \sin(2t), \cos(t) - 2 \cos(2t), \sin(3t))$$

and

$$((2 + \cos(3t)) \cos(2t), (2 + \cos(3t)) \sin(2t), \sin(3t))$$

for $t \in [0, 2\pi)$.

Writing $t = t(\theta)$, the magnitude of the derivatives wrt. θ of these two parameterizations are:

$$\begin{aligned} & \left| \frac{d}{d\theta} (\sin(t) + 2 \sin(2t), \cos(t) - 2 \cos(2t), \sin(3t)) \right| \\ &= \left| (\cos(t) + 4 \cos(2t), -\sin(t) + 4 \sin(2t), 3 \cos(3t)) \frac{dt}{d\theta} \right| \\ &= \left(\cos^2(t) + \sin^2(t) + 16[\cos^2(2t) + \sin^2(2t)] \right. \\ &\quad \left. + 8[\cos(t) \cos(2t) - \sin(t) \sin(2t)] + 9 \cos^2(3t) \right)^{1/2} \frac{dt}{d\theta} \\ &= \sqrt{17 + 8 \cos(3t) + 9 \cos^2(3t)} \frac{dt}{d\theta} \end{aligned}$$

and similarly

$$\begin{aligned} & \left| \frac{d}{d\theta} ((2 + \cos(3t)) \cos(2t), (2 + \cos(3t)) \sin(2t), \sin(3t)) \right| \\ &= \sqrt{25 + 16 \cos(3t) + 4 \cos^2(3t)} \frac{dt}{d\theta} \end{aligned}$$

I need a solution to, for example:

$$\int \sqrt{17 + 8 \cos(3t) + 9 \cos^2(3t)} dt$$

so that I can get t in terms of θ .

Wolfram Alpha doesn't like either of these integrals, and I can see no way to solve them. There are two types of answers to this question: one would solve one of these integrals, another would tell me how to change the parameterization, reasonably, so that the integral, and the obtained formula for t , is solvable. I guess the third is to tell me that this question isn't solvable like this.

My thoughts on the latter: Below, I make the integral solvable by changing the parameterization, but I can't solve for $t(\theta)$.

The freedom in the parameterizations can be expressed as:

$$(\sin(t) + A \sin(2t), \cos(t) - A \cos(2t), B \sin(3t))$$

where $A > 1$ and $B > 0$, which gives integrand:

$$\sqrt{1 + 4A^2 + 4A \cos(3t) + 9B^2 \cos^2(3t)}$$

To eliminate the square root we want some factorization:

$$(1 + 4A^2 + 4A \cos(3t) + 9B^2 \cos^2(3t)) = (3B \cos(3t) + \lambda)^2$$

where $\lambda^2 = 1 + 4A^2$ and $6B\lambda = 4A$, so $\lambda = \frac{2A}{3B}$.

So $\frac{4A^2}{9B^2} = 1 + 4A^2$, i.e. $4A^2 = \frac{9B^2}{1-9B^2} > 4$. So $9B^2 > \frac{4}{5}$ and I will choose $9B^2 = \frac{9}{10}$ giving:

$$B = \frac{1}{\sqrt{10}}, \quad A = \frac{3}{2}, \quad \lambda = \sqrt{10}$$

i.e. the trefoil $(\sin(t) + \frac{3}{2} \sin(2t), \cos(t) - \frac{3}{2} \cos(2t), \frac{1}{\sqrt{10}} \sin(3t))$ has arclength parameterization given by $t(\theta)$ where:

$$\int \left(\frac{3}{\sqrt{10}} \cos(3t) + \sqrt{10} \right) dt = \theta$$

$$\frac{\sin(3t)}{\sqrt{10}} + t\sqrt{10} = \theta$$

But I want t in terms of θ .

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